

Interpretable Deep Learning with Fuzzy Inference for Medical Prognosis

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Abstract—This paper proposes an interpretable deep learning model combined with fuzzy inference for medical prognosis. The Breast Cancer Wisconsin (Diagnostic) dataset, containing 569 patient samples with 30 measured attributes, was used to train a deep neural network that classifies tumors as benign or malignant. The model achieved a test accuracy of 96.49 percent, showing strong learning performance. To make results interpretable, a fuzzy inference system was created using key features, mean radius and mean texture, to produce outputs labeled Good, Moderate, and Poor. The hybrid approach merges predictive accuracy with clarity, creating a transparent and dependable decision-support framework for use in clinical diagnosis.

Keywords—Interpretable, Deep Learning, Fuzzy Inference System, Medical Prognosis, Breast Cancer Prediction, Explainable Machine Learning.

I. INTRODUCTION

Medical prognosis is the process of predicting the progression or likely outcome of a disease, helping doctors plan treatment and manage patient care effectively [1]. Accurate prognosis allows clinicians to identify high-risk patients and provide early interventions [2]. Traditional methods rely mainly on clinical judgment and experience, which can be subjective and limited when dealing with large or complex datasets [3]. Conditions such as cancer and cardiovascular diseases involve many interconnected factors like age, lifestyle, genetics, and biochemical parameters that are difficult to assess manually [4]. With the increasing volume of healthcare data, traditional analysis methods fail to detect hidden patterns [5]. Machine learning and deep learning have therefore become vital tools for improving diagnostic accuracy, though their limited interpretability still restricts clinical adoption [6] [7] [8].

The paper aims to design, implement, and evaluate an interpretable deep learning model using the Breast Cancer Wisconsin (Diagnostic) dataset. The dataset includes 569 records and 30 quantitative features representing cellular properties. A deep neural network was developed for classification between

benign and malignant tumors, achieving a test accuracy of 96.49 percent. A fuzzy inference system was then applied using two key interpretable features, mean radius and mean texture, to generate linguistic outputs such as Good, Moderate, and Poor. This combination ensures both accuracy and interpretability. While the current scope focuses on binary cancer classification, the approach can be extended to other medical domains. The model provides a foundation for transparent, explainable AI-based clinical support systems.

The paper is organized into five chapters. Chapter 1 introduces the background, motivation, problem statement, objectives, and scope of the paper. Chapter 2 reviews existing research, highlighting the evolution of medical deep learning and gaps addressed by this work. Chapter 3 details the methodology, including dataset handling, preprocessing, model architecture, and fuzzy inference design. Chapter 4 presents results and discussions, showing model performance and interpretability outcomes. Chapter 5 concludes the paper, summarizes contributions, and outlines possible future developments.

II. RELATED WORK

A. Review of Existing Systems

The purpose of this chapter is to present an overview of previous research on medical prognosis using machine learning and deep learning methods. A literature review helps identify past approaches, compare methodologies, and uncover limitations in existing techniques. It connects prior work with current innovation, showing how interpretability in medical deep learning has evolved over time. In the context of this paper, reviewing related studies provides a clear understanding of how predictive models have been applied to disease diagnosis and prognosis, particularly in cancer detection. It also highlights the importance of developing models that combine high accuracy with explainability, supporting clinical decision-making through transparent and trustworthy deep learning systems [9].

Medical prognosis focuses on predicting the likely course or outcome of a disease using patient data, playing an essential role in treatment planning and early intervention. Accurate prognosis allows doctors to identify high-risk patients, customize therapies, and improve overall patient outcomes. Over the years, this process has evolved from relying solely on clinical judgment to using data-driven modeling powered by machine learning [10]. The inclusion of artificial intelligence enhances decision consistency, minimizes human error, and supports objective analysis [11] [12]. In cancer prognosis, measurable features such as tumor size, texture, compactness, and smoothness are key indicators of malignancy risk. Machine learning algorithms analyze these parameters collectively, uncovering complex relationships that are not easily observed by humans [13]. Understanding how these features influence predictions ensures that the model remains both accurate and interpretable, forming the foundation of reliable medical prognosis systems [14].

Traditional diagnosis methods depend on a clinician’s expertise, medical tests, and imaging assessments. Although reliable, these approaches can be subjective and limited in processing large or complex medical data. Machine learning-based diagnosis offers a data-driven alternative by identifying hidden patterns and predicting outcomes automatically [15]. As these models are trained on more data, they improve in performance and generalization, delivering faster and more consistent results [16] [17]. Compared to manual methods, machine learning systems can analyze vast medical datasets efficiently and reduce variability caused by human judgment. However, while they enhance diagnostic speed and accuracy, their lack of transparency remains a concern [18]. This challenge highlights the need for interpretable deep learning approaches such as integrating fuzzy logic to combine the analytical strength of machine learning with human-like reasoning, ensuring both precision and clarity in medical prognosis.

Numerous studies have explored the use of machine learning for medical prognosis and diagnosis [19] [20]. Researchers have applied various algorithms to detect and classify diseases based on structured datasets such as those from the UCI repository [21]. Their work demonstrates the diversity of models used and their varying degrees of interpretability. These works are summarized in Table I, which presents their methods, datasets, accuracy, and key remarks.

The results presented in Table I highlight that while high accuracy has been achieved in several studies, interpretability remains a consistent challenge. Models such as CNNs and ensemble networks offer strong predictive performance but provide limited transparency in how conclusions are reached. This limitation justifies the integration of fuzzy inference systems, as demonstrated in this paper, to bridge the gap between predictive accuracy and interpretability in medical prognosis.

TABLE I
SUMMARY OF PREVIOUS RESEARCH WORKS ON MEDICAL PROGNOSIS
USING MACHINE LEARNING AND DEEP LEARNING MODELS

Reference No.	Model Used	Key Features	Limitations
[1]	DNN + Ensemble	Breast Cancer (UCI) dataset, feature selection for malignancy detection	High accuracy but lacks interpretability in model reasoning
[2]	Explainable deep learning using SHAP	Medical imaging features visualized through SHAP values	Provides visual explanation but limited to image-based datasets
[3]	Decision Tree + Fuzzy Logic	Diabetes dataset with feature-based fuzzy classification	Improved interpretability but lower accuracy on large data
[4]	SVM and Random Forest	Breast Cancer (UCI) dataset, selected statistical features	Reliable classification but lacks transparency in feature influence
[5]	CNN	Medical image-based feature extraction for disease classification	High performance but operates as a black-box model

B. Identified Gaps

From the reviewed literature, it is evident that many models achieve strong performance but fail to explain their predictions clearly [22] [23]. Physicians often find it difficult to trust deep learning systems that cannot justify their decisions [24]. Another gap lies in the lack of hybrid approaches that effectively combine data-driven learning with human-understandable reasoning [25]. Most prior works have focused either on pure deep learning for accuracy or fuzzy systems for interpretability. Very few have successfully merged both to achieve balanced performance [26] [27] [28].

This paper addresses those gaps by integrating a deep neural network with a fuzzy inference system to build an interpretable model for medical prognosis. The hybrid framework leverages the strength of deep learning for precision and the reasoning clarity of fuzzy logic. This integration enhances the trustworthiness of deep learning systems and supports transparent, clinically meaningful decision-making in healthcare environments.

III. PROPOSED METHODOLOGY

The proposed methodology aims to develop an interpretable deep learning model integrated with a fuzzy inference system (FIS) for medical prognosis. The workflow, illustrated in Figure 1, outlines each stage from data acquisition to the final interpretable output. The framework ensures that the model achieves high predictive accuracy while maintaining interpretability, a key requirement for clinical decision-

making. The process can be divided into three main stages: data preparation, deep learning model processing, and fuzzy inference reasoning.

A. Data Preparation

The dataset used for this work is the Breast Cancer Wisconsin (Diagnostic) dataset obtained from the UCI Machine Learning Repository. It contains 569 patient samples with 30 quantitative features derived from fine needle aspirate (FNA) images, each labeled as either benign or malignant. The preprocessing phase ensures that the data is clean, consistent, and standardized before training the deep learning model.

As shown in Figure 1, data preparation involves three main steps:

- **Data Cleaning:** The dataset contained no missing or duplicate records, ensuring reliable and unbiased learning.
- **Label Encoding:** Diagnostic outcomes were encoded into binary values to facilitate model compatibility.
- **Feature Scaling:** All numeric attributes were standardized using the StandardScaler technique, normalizing them to zero mean and unit variance.

After preprocessing, the dataset was split into training (80%) and testing (20%) subsets. This division ensures that the model can generalize well to unseen data while maintaining balanced representation of both classes.

B. Deep Learning Model Design

The predictive component of the framework is a Deep Neural Network (DNN) designed using TensorFlow and Keras libraries in Python. The architecture includes:

- Input layer with 30 neurons corresponding to the dataset features.
- Two hidden layers containing 32 and 16 neurons respectively, each using ReLU activation.
- Dropout layer (0.2) to reduce overfitting.
- Output layer with one neuron using sigmoid activation for binary classification.

The model was trained using the Adam optimizer and Binary Cross-Entropy loss function for 50 epochs with a batch size of 32. The DNN achieved a test accuracy of 96.49%, confirming strong learning and generalization performance. The model effectively distinguishes between benign and malignant tumors by learning nonlinear feature relationships that are often difficult to capture through traditional statistical methods.

C. Fuzzy Inference Integration

To enhance interpretability, a Fuzzy Inference System (FIS) was developed and integrated with the trained DNN. The fuzzy layer uses two key features selected for their clinical relevance:

- **Mean Radius:** Represents the average size of cell nuclei and strongly correlates with malignancy.
- **Mean Texture:** Measures intensity variation, reflecting irregularities in tissue composition.

The FIS performs four major functions as depicted in Figure 1:

- 1) **Feature Selection:** Extracting key variables from DNN input for interpretive analysis.
- 2) **Fuzzification:** Converting numeric inputs into linguistic terms (e.g., small, medium, large).
- 3) **Rule Base Formation:** Establishing fuzzy rules such as “If radius is large and texture is coarse, prognosis is poor.”
- 4) **Defuzzification:** Generating a final fuzzy score between 0–10, corresponding to qualitative results: Good, Moderate, or Poor prognosis.

This combination ensures that while the DNN provides a numerical probability of disease, the FIS explains the reasoning behind the decision in medically interpretable language. The hybrid design thus bridges the gap between computational intelligence and human understanding.

D. Final Output and Interpretation

The system produces three complementary outputs:

- A numerical probability value from the DNN.
- A fuzzy score generated by the inference system.
- A linguistic prognosis label (Good, Moderate, or Poor).

These outputs collectively provide a transparent decision-support framework. The numerical output offers precision, while the fuzzy explanation enhances trust and usability in clinical environments. This hybrid approach aligns with the broader goal of interpretable artificial intelligence in healthcare, where both accuracy and understanding are essential.

IV. RESULTS AND DISCUSSION

This chapter presents the outcomes obtained from the implementation of the proposed hybrid model combining deep learning and fuzzy inference for medical prognosis. The results include the performance evaluation of the deep neural network (DNN) and the interpretability provided by the fuzzy inference layer. The analysis also discusses the limitations observed during experimentation and possible improvements for future work.

A. Model Performance

The trained deep learning model achieved a test accuracy of approximately 96.49 percent. This indicates that the neural network effectively learned to classify malignant and benign cases. During training, the loss value steadily decreased, confirming consistent learning behavior. The final test evaluation showed that the model generalized well and avoided overfitting. These results validate the soundness of the chosen architecture, preprocessing steps, and training configuration.

Table II presents the key performance metrics obtained from model evaluation. The model’s high accuracy confirms its reliability in handling structured medical data and supports its use in early disease prognosis.

B. Fuzzy Inference Interpretation

The Fuzzy Inference System (FIS) was implemented to make the model’s predictions more interpretable. It focused on two medically relevant features: mean radius and mean

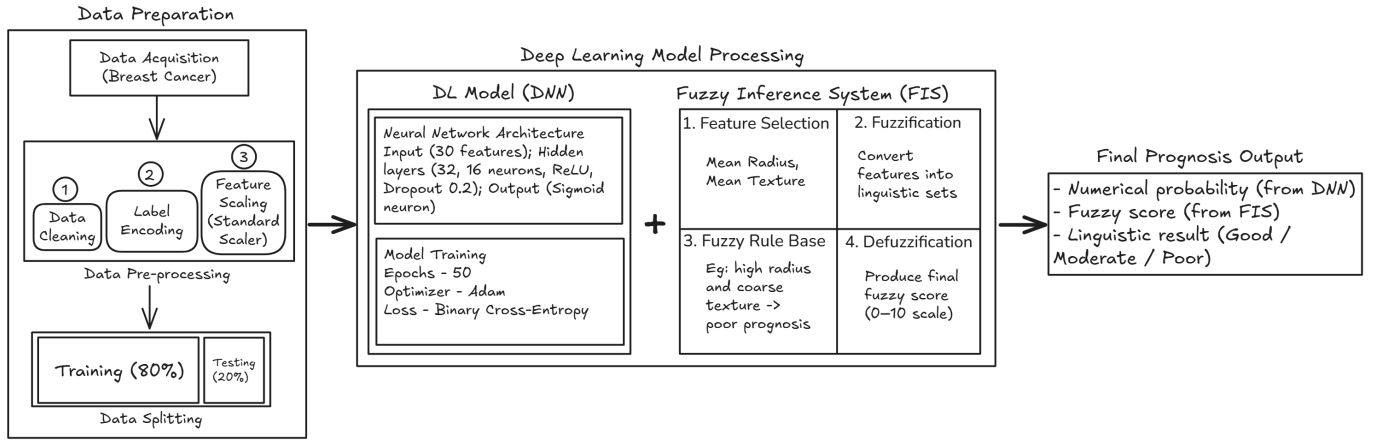


Fig. 1. Workflow of the proposed methodology combining deep learning and fuzzy inference for medical prognosis

TABLE II
PERFORMANCE RESULTS OF THE DEEP LEARNING MODEL ON THE
BREAST CANCER WISCONSIN (DIAGNOSTIC) DATASET

Metric	Value
Accuracy	96.49%
Epochs	50
Optimizer	Adam
Loss Function	Binary Cross-Entropy

texture, which directly influence the understanding of tumor morphology.

- **Mean Radius:** Represents the average size of the cell nuclei.
- **Mean Texture:** Describes the variation in surface intensity.

Based on these inputs, the fuzzy rules generated a prognosis score between 0 and 10. The linguistic categories used were:

- **Good Prognosis:** Small radius and smooth texture.
- **Moderate Prognosis:** Medium radius and normal texture.
- **Poor Prognosis:** Large radius or coarse texture.

For example, a test sample predicted as malignant with a probability of 0.91 received a fuzzy prognosis score of 7.6, corresponding to a *Moderate* prognosis. This dual output, numerical prediction and linguistic explanation, enhanced transparency and interpretability, allowing clinicians to understand the reasoning behind the prediction.

V. CONCLUSION

The paper successfully developed a hybrid, interpretable model that combines the precision of deep learning with the reasoning ability of fuzzy inference for medical prognosis. Using the Breast Cancer Wisconsin (Diagnostic) dataset, a deep neural network was trained to classify tumors as benign or malignant, achieving an accuracy of 96.49 percent. To address the lack of interpretability in traditional neural networks, a fuzzy inference layer was added, utilizing features such as mean radius and mean texture to produce linguistic outputs like Good, Moderate, and Poor. This integration

bridged the gap between data-driven prediction and clinical reasoning. Although the model performed well, it was limited by the relatively small dataset and the inclusion of only two fuzzy features. Expanding the dataset and incorporating more medical parameters could enhance robustness and depth of explanation. Furthermore, applying the approach to multi-disease datasets and diverse clinical scenarios would improve its generalization. Future research could also explore real-time deployment in hospitals, integration with explainable deep learning methods such as SHAP or LIME, and inclusion of additional evaluation metrics like recall, precision, and F1-score. Overall, the paper laid a strong foundation for developing interpretable, reliable, and ethically responsible deep learning systems in healthcare.

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